

AI VIET NAM – AI Conquer 2026

Research Report – Module 1

Similarity Matching Methods for Images and Texts

Name 1, Name 2, Name 3, Name 4 & Name 5

1 Abstract

This report studies similarity matching methods through two complementary modules: text-to-text matching and image-to-image matching. The text-to-text module investigates how textual inputs can be compared using classical surface-matching methods, distributed corpus embeddings, and modern sentence-vector methods based on large language models. The image-to-image module evaluates how images can be compared through hand-crafted descriptors, local feature matching, CNN embeddings, metric learning, and foundation-model embeddings. Both modules are organized under the same experimental logic: represent each input as a numerical form, calculate similarity, rank candidate matches, and evaluate the quality of the ranking. The report emphasizes conceptual understanding, practical implementation, and comparative analysis under limited computing resources.

List of Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
LCS	Longest Common Subsequence	TF-IDF	Term Frequency - Inverse Document Frequency
CBOW	Continuous Bag-of-Words	GloVe	Global Vectors for Word Representation
LSTM	Long Short-Term Memory	MaLSTM	Manhattan LSTM
RCNN	Recurrent/Region-based Convolutional Neural Network	DSSTM	Deep Structured Semantic Model
DRCN	Deep Residual Convolutional Neural Network	BiMPM	Bilateral Multi-Perspective Matching
LLMs	Large Language Models	BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Pretraining Approach	GPT	Generative Pre-trained Transformer
R2-Net	Relation of Relation Network	DC-Match	Dual-Context / Divide-and-Conquer Matching
BoVW	Bag-of-Visual-Words	CNN	Convolutional Neural Network
CLIP	Contrastive Language-Image Pretraining	DINO	Self-supervised Vision Transformer model
F1@K	F1-score at top K retrieved results	GeM	Generalized Mean Pooling
GIST	Global image scene descriptor	HOG	Histogram of Oriented Gradients
LBP	Local Binary Pattern	mAP	Mean Average Precision
MSE	Mean Squared Error	ORB	Oriented FAST and Rotated BRIEF
P@K	Precision at top K retrieved results	R@K	Recall at top K retrieved results
RAM	Random Access Memory	RGB / HSV	Color spaces used for image features
SIFT	Scale-Invariant Feature Transform	SigLIP	Sigmoid Loss for Language-Image Pretraining
SSIM	Structural Similarity Index Measure	VGG	Visual Geometry Group CNN family

Contents

1 Abstract	1
List of Abbreviations	2
2 Introduction	5
A. Text-to-Text Matching	5
B. Image-to-Image Matching	5
3 Problem Statement	5
A. Text-to-Text Matching	5
B. Image-to-Image Matching	5
4 Dataset	6
A. Text-to-Text Matching	6
B. Image-to-Image Matching	6
5 Methodology	6
A. Text-to-Text Matching	6
B. Image-to-Image Matching	7
6 System Implementation	8
A. Text-to-Text Matching	8
B. Image-to-Image Matching	8
7 Experiments and Evaluation	9
A. Text-to-Text Matching	9
B. Image-to-Image Matching	10
8 Error Analysis and Discussion	12
A. Text-to-Text Matching	12
B. Image-to-Image Matching	12
9 Conclusion and Future Work	13
A. Text-to-Text Matching	13
B. Image-to-Image Matching	13

10	References	14
A	Appendix	15
A.	Text-to-Text Matching Supplement	15
B.	Image-to-Image Matching Supplement	15

2 Introduction

A. Text-to-Text Matching

Text-to-text similarity matching measures how close two textual inputs are in meaning, form, or context. It is useful for semantic search, duplicate question detection, information retrieval, plagiarism checking, and retrieval-augmented generation systems. Classical methods compare characters, tokens, or word overlap, while modern methods encode sentences into dense vectors and compare their semantic positions in an embedding space.

The motivation for this module is that exact keyword matching is often too shallow. Two sentences may use different words but express the same idea, while two sentences with many shared words may differ in meaning. Therefore, text similarity matching requires both surface-level and semantic-level representations.

B. Image-to-Image Matching

Image-to-image similarity matching measures whether two images are visually, structurally, or semantically related. It supports applications such as landmark retrieval, duplicate image detection, visual search, product search, and content-based image retrieval. Classical methods compare pixels, color distributions, edges, or local keypoints. Modern methods compare deep embeddings extracted from CNNs, Siamese networks, or foundation models.

The motivation for this module is that raw pixel comparison is usually not enough. Two images of the same building may differ because of viewpoint, crop, lighting, scale, and background clutter. A practical image retrieval system must therefore compare images using representations that are robust to visual changes.

3 Problem Statement

A. Text-to-Text Matching

Given a pair of text inputs (s_1, s_2) , the goal is to estimate their semantic similarity. The expected output is a continuous similarity score or ranking relationship that reflects how close the model prediction is to a human similarity judgment. In the STS Benchmark setting, this is treated as a regression-style similarity evaluation problem. The primary criteria are Pearson correlation and Spearman correlation between predicted similarity scores and human labels.

B. Image-to-Image Matching

Given a query image and a database of candidate images, the goal is to rank database images from most similar to least similar. The expected output is an ordered retrieval list. For CIFAR-10, relevance is defined at the class level: images from the same class are considered relevant. For Oxford5k/Paris6k, relevance is defined at the landmark-instance level: images should show the same building or landmark. The primary evaluation criteria include mAP, Precision@K, Recall@K, F1@K, Top-1 Accuracy, inference latency, and memory usage.

4 Dataset

A. Text-to-Text Matching

The text-to-text experiment uses the STS Benchmark (STSb). It contains approximately 8,600 sentence pairs, each annotated with a continuous human similarity score from 0.0 to 5.0. A score near 0.0 indicates that the sentences are unrelated, while a score near 5.0 indicates that they are nearly equivalent in meaning. This makes STSb suitable for evaluating semantic similarity as a regression-style task.

B. Image-to-Image Matching

The image-to-image experiments use two retrieval settings. CIFAR-10 is used as a lightweight category-level benchmark. Each image belongs to one of ten classes, and images from the same class are treated as relevant. Oxford5k and Paris6k are used as landmark retrieval datasets. They evaluate whether a method can retrieve images of the same building or landmark under changes in viewpoint, lighting, scale, crop, and background.

5 Methodology

A. Text-to-Text Matching

Text similarity methods can be divided into seven groups:

1. **String-based methods:** methods based on character or phrase overlap, such as Longest Common Subsequence, Edit Distance, Dice coefficient, and Jaccard Similarity.
2. **Corpus-based methods:** methods based on word statistics or distributed representations, such as TF-IDF, CBOW, Word2Vec, GloVe, and Doc2Vec.
3. **Recursive neural network matching:** recurrent or recursive models such as MaLSTM, Siamese LSTM, and Enhanced-RCNN.
4. **Sentence semantic interaction methods:** models such as DSSTM, DRCN, and BiMPM that compare two sentences through richer interaction mechanisms.
5. **Graph-structure methods:** approaches such as Wpath and Graph Matching Networks that use graph-based semantic or structural information.
6. **Sentence-vector methods based on LLMs:** BERT, RoBERTa, GPT-family models, and other large language models used to produce contextual sentence vectors.
7. **LLM matching with different features:** hybrid models such as R2-Net, ASPIRE, and DC-Match that combine multiple textual features.

For the practical experiment, two representative approaches are emphasized: distributed corpus embeddings and sentence-vector matching based on LLMs.

Table 1: Two representative approaches for text similarity matching.

Tên phương pháp	Distributed Corpus Matching Method	Sentence Vector-Matching based on LLMs
Mô hình đại diện	Doc2Vec hoặc Averaged GloVe	Bi-Encoder, ví dụ all-MiniLM-L6-v2
Đặc trưng kỹ thuật	Static embeddings: mỗi từ hoặc văn bản có một vector cố định, không thay đổi theo ngữ cảnh.	Contextualized embeddings: dùng cơ chế Attention của Transformer để hiểu ý nghĩa từ theo ngữ cảnh xung quanh.
Ưu điểm trên laptop	Nhẹ, yêu cầu ít RAM, tốc độ huấn luyện và trích xuất vector nhanh trên CPU.	Kích thước gọn, xử lý nhanh trên CPU nhưng cho độ chính xác cao hơn nhờ pretraining hiện đại.

B. Image-to-Image Matching

Image similarity methods are grouped from classical to modern approaches:

1. **Pixel-based and structural methods:** MSE, SSIM, and template matching compare images at the pixel or structure level.
2. **Color and texture-based methods:** Color Histogram, LBP, and Gabor filters capture low-level visual statistics.
3. **Local feature matching methods:** SIFT, SURF, ORB, BRIEF, and AKAZE detect and compare local keypoints.
4. **Global descriptor methods:** HOG, GIST, BoVW, VLAD, and Fisher Vectors summarize the whole image into a vector.
5. **Deep CNN embedding methods:** ResNet, VGG, EfficientNet, and ConvNeXt extract learned visual embeddings.
6. **Metric learning methods:** Siamese networks, contrastive loss, and triplet loss train embeddings directly for similarity.
7. **Self-supervised and vision-language foundation models:** SimCLR, MoCo, BYOL, DINO, CLIP, and SigLIP use large-scale pretraining to learn transferable visual representations.

The implemented image methods are organized into three groups: hand-crafted global and structural methods, local feature and visual-word methods, and deep embedding methods. This allows the experiment to compare different representation levels: pixel/structure-level, local-feature-level, and learned semantic embedding-level.

Table 2: Comparison between classical and modern image similarity methods.

Aspect	Classical hand-crafted methods	Modern deep embedding methods
Core idea	Compare manually designed visual features such as color, texture, edges, structure, or keypoints.	Compare learned embeddings from pretrained CNNs, Siamese networks, or foundation models.
Representation	Low-level or mid-level descriptors such as histograms, HOG vectors, SIFT descriptors, or BoVW histograms.	Dense semantic vectors learned from large-scale image or image-text pretraining.
Robustness	Sensitive to lighting, viewpoint, object scale, and background changes.	More robust to semantic and visual variation, especially when pretrained on large datasets.
Interpretability	Easier to explain because features are human-designed.	Harder to interpret because embeddings are learned internally.
Computation	Usually lightweight and CPU-friendly.	More computationally expensive and often benefits from GPU acceleration.

6 System Implementation

A. Text-to-Text Matching

The text-to-text system follows a three-step pipeline. First, each sentence is encoded into a vector representation using either a classical embedding method or a sentence-vector model. Second, cosine similarity is applied to compare the two vectors. Third, the predicted similarity scores are compared with human labels using Pearson and Spearman correlation.

The main design decision is to use a bi-encoder style pipeline for modern sentence embeddings. This is efficient because each sentence can be encoded independently, and similarity can be computed using vector operations. It is lighter than cross-encoder architectures, which compare sentence pairs jointly and are usually more expensive.

B. Image-to-Image Matching

The image-to-image system treats retrieval as a ranking problem. Each method first extracts a visual representation from the query and database images. Global vector methods and deep embeddings are compared using cosine similarity. SSIM uses a structural similarity score. SIFT uses descriptor matching with Lowe’s ratio test. BoVW builds visual-word histograms from SIFT descriptors and compares the resulting histograms.

For deep embedding methods, images are resized and normalized according to the pretrained model requirements. CNN backbone variants use ResNet50, EfficientNet-B0, and VGG19 with either average pooling or GeM pooling. The final retrieval list is produced by sorting database images according to similarity score.

7 Experiments and Evaluation

A. Text-to-Text Matching

The text-to-text experiment evaluates how well predicted similarity scores match human similarity judgments on STSb. The main metrics are Spearman correlation and Pearson correlation. Spearman correlation measures whether the model ranks sentence pairs in a similar order to human judgments, while Pearson correlation measures whether the predicted score values align linearly with the ground-truth labels.

Table 3: Comparison between classical and modern text similarity methods.

Đặc điểm so sánh	Nhóm phương pháp cổ điển	Nhóm phương pháp hiện đại
Bản chất cốt lõi	Khớp từ khóa bề mặt, tần suất từ hoặc biểu diễn vector tĩnh.	Hiểu ngữ nghĩa ẩn, bối cảnh từ trong câu và tương tác không gian sâu.
Khả năng hiểu từ đồng nghĩa	Rất kém hoặc không thể nếu chỉ dựa trên chuỗi/từ khóa. Ví dụ “máy tính” và “computer” có thể bị xem như khác nhau hoàn toàn.	Tốt hơn vì các từ/câu gần nghĩa có thể nằm gần nhau trong không gian vector ngữ nghĩa.
Độ nhạy với trật tự từ	Kém với các mô hình Bag-of-Words như TF-IDF, vì chúng thường bỏ qua thứ tự từ.	Tốt hơn nhờ RNN, Attention hoặc positional encoding giúp mô hình giữ thông tin thứ tự và ngữ cảnh.
Yêu cầu phần cứng	Thấp, chạy tốt trên CPU, tốn ít RAM.	Cao hơn, đặc biệt khi huấn luyện hoặc dùng Cross-Encoder.
Tốc độ xử lý	Nhanh vì chủ yếu dùng công thức thống kê hoặc so khớp rời rạc.	Chậm hơn, phụ thuộc vào số tham số và kiến trúc mô hình.

Table 4: Practical comparison between distributed corpus embeddings and LLM sentence vectors.

Đặc điểm đối chiếu	Distributed Corpus (Word2Vec, GloVe, Doc2Vec)	Sentence Vector-Matching based on LLMs
Tính đa nghĩa của từ	Vector tĩnh: một từ chỉ có một vector cố định. Ví dụ “bank” trong “river bank” và “commercial bank” có thể dùng cùng vector, gây sai lệch ngữ nghĩa.	Vector động: biểu diễn của từ thay đổi theo ngữ cảnh, nên cùng một từ có thể có vị trí vector khác nhau trong hai câu khác nhau.
Phương thức tạo vector câu	Thường lấy trung bình hoặc tổng vector từ, dễ làm loãng ý nghĩa và mất cấu trúc cú pháp.	Dùng pooling trên hidden states hoặc attention để tạo vector câu chất lượng cao hơn.
Khả năng fine-tune theo domain	Hạn chế; thường phải huấn luyện lại hoặc khó giữ tri thức cũ.	Tốt hơn; có thể fine-tune pretrained model với dữ liệu nhỏ bằng các loss phù hợp như CosineSimilarityLoss.
Độ chính xác thực tế trên STSb	Trung bình thấp, Spearman thường quanh 0.40–0.45.	Cao hơn rõ rệt; MiniLM có thể đạt khoảng 0.84 Spearman trong thực nghiệm.

B. Image-to-Image Matching

The image-to-image experiments evaluate seven implemented methods on CIFAR-10 and Oxford5k/Paris6k: Color Histogram + HOG, SSIM, SIFT with Lowe’s ratio test, BoVW based on SIFT descriptors, EfficientNet transfer-learning embeddings, Siamese Network with an EfficientNet backbone, and SigLIP embeddings. Higher values are better for mAP, Precision@K, Recall@K, F1@K, and Top-1 Accuracy. Lower values are better for latency and memory usage.

Table 5: Cross-dataset comparison for image-to-image retrieval.

Method	CIFAR-10 mAP	CIFAR-10 R@10	Oxford/Paris mAP	Oxford/Paris R@10
Color Histogram	0.1573	0.0008	0.0338	0.0065
SSIM	0.1274	0.0006	0.0313	0.0035
SIFT	0.0211	0.0000	0.0000	0.0000
BoVW	0.1201	0.0004	0.0614	0.0172
EfficientNet	0.4741	0.0017	0.3946	0.0836
Siamese	0.6068	0.0014	0.4311	0.0690
SigLIP	0.6344	0.0018	0.4897	0.1183

Table 6: Evaluation on Oxford5k/Paris6k.

Method	mAP	P@10	R@10	F1@10	Top-1 Acc.	Latency / Query (ms)	Memory (MB)
Color Histogram	0.0338	0.7214	0.0065	0.0121	0.2857	1458	1363
SSIM	0.0313	0.0829	0.0035	0.0065	0.1429	13291	1549
SIFT	0.0211	0.0000	0.0000	0.0000	0.0000	55616	2783
BoVW	0.0614	0.1921	0.0172	0.0296	0.2929	238	3310
EfficientNet	0.3946	0.6386	0.0836	0.1071	0.7786	1567	4832
Siamese	0.4311	0.6357	0.0690	0.1105	0.6929	3303	5535
SigLIP	0.4897	0.7214	0.1183	0.1341	0.8286	2205	6985

Table 7: Comparison of CNN backbones on Oxford5k/Paris6k.

Method	mAP	P@10	R@10	F1@10	Top-1 Acc.	Latency / Query (ms)	Memory (MB)
ResNet50 AvgPool	0.3573	0.5950	0.0555	0.0839	0.6429	1415	2763
ResNet50 GeM	0.3268	0.5914	0.0541	0.0826	0.6357	1377	2727
EfficientNet AvgPool	0.3946	0.6386	0.0836	0.1071	0.7786	1524	2490
EfficientNet GeM	0.3909	0.6271	0.0862	0.1033	0.7571	1551	2530
VGG19 AvgPool	0.2958	0.5814	0.0629	0.0889	0.6500	1618	6730
VGG19 GeM	0.2959	0.5943	0.0615	0.0882	0.6857	1543	6748

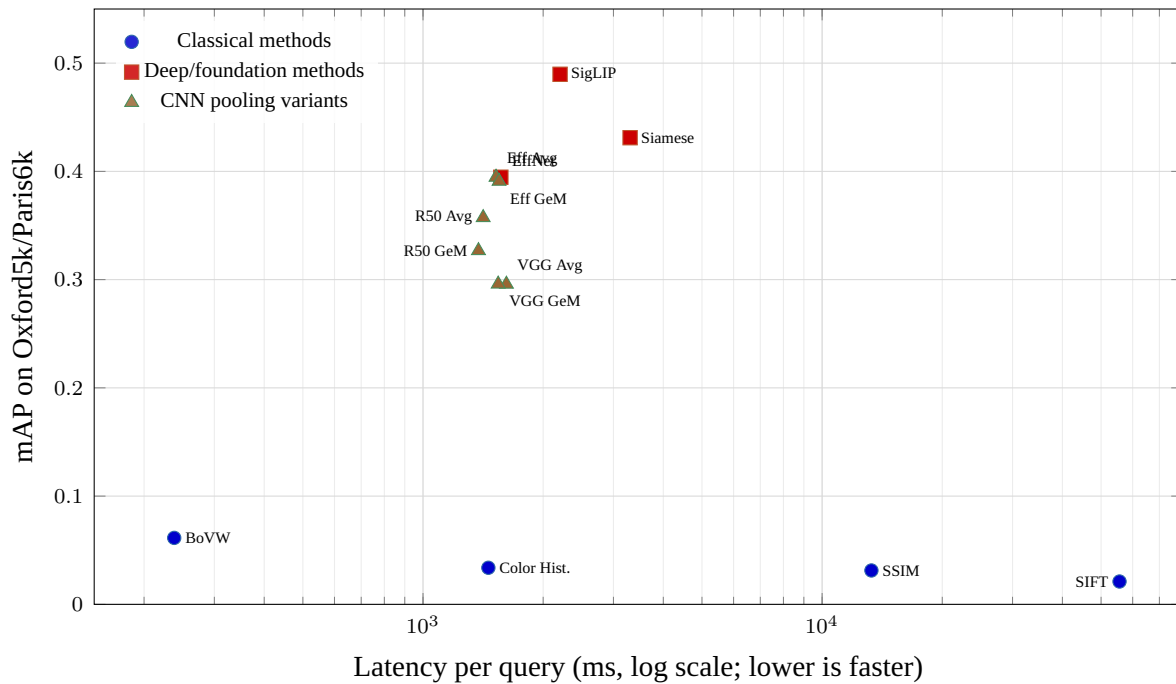


Figure 1: Speed-accuracy trade-off across all implemented image methods.

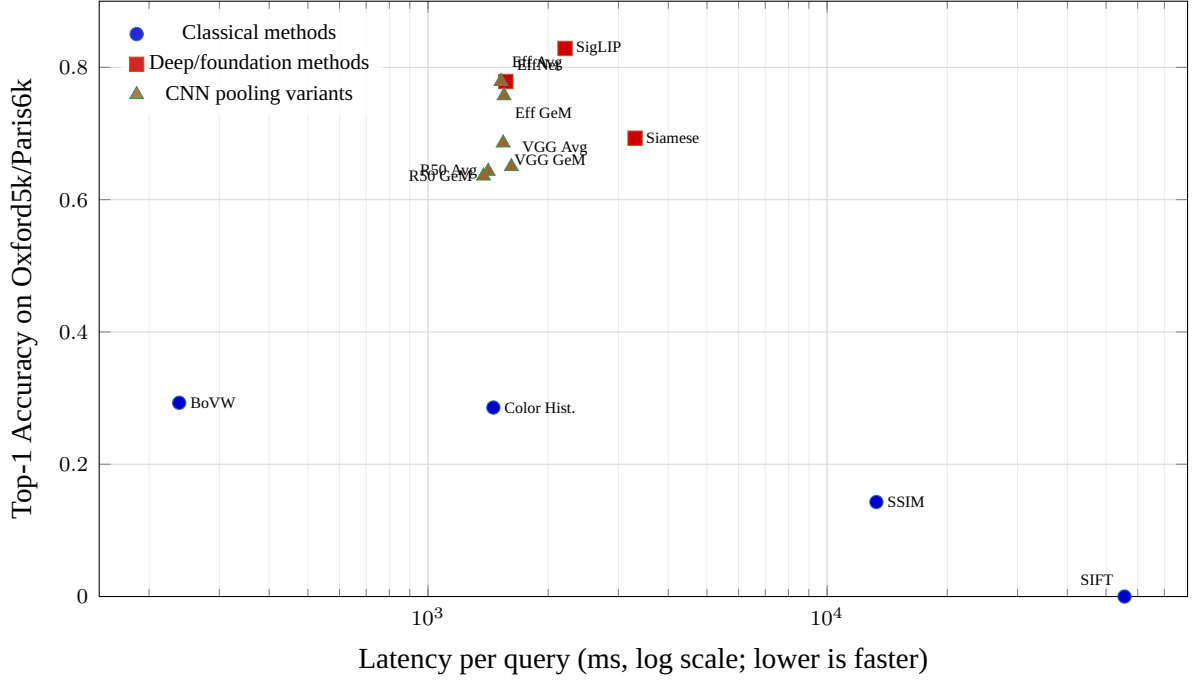


Figure 2: Speed versus Top-1 Accuracy across all implemented image methods.

8 Error Analysis and Discussion

A. Text-to-Text Matching

The main limitation of classical text methods is that they focus heavily on surface form. String-based methods are useful for spelling errors, short titles, or exact duplicate detection, but they cannot reliably detect synonymy or paraphrase. Corpus-based embeddings improve this by creating a vector space, but older distributed embeddings are usually static: the same word keeps the same vector even when its meaning changes by context.

Modern sentence-vector models address this limitation by creating contextualized embeddings. However, they are more computationally expensive and may still fail when the sentence requires domain knowledge, numerical reasoning, or subtle pragmatic interpretation. The development from Word2Vec and GloVe to LLM sentence embeddings can be understood as a shift from static vector spaces to contextualized representations. Word2Vec/GloVe introduced the idea that language can be represented geometrically, while CBOW/Skip-gram inspired later pretraining objectives such as Masked Language Modeling and Causal Language Modeling.

B. Image-to-Image Matching

The image retrieval results show three main observations. First, deep embedding methods clearly outperform hand-crafted methods on both CIFAR-10 and Oxford5k/Paris6k. EfficientNet, Siamese EfficientNet, and SigLIP achieve much higher mAP than Color Histogram + HOG, SSIM, SIFT, and BoVW. This suggests that learned visual representations capture semantic and instance-level similarity better than manually designed low-level features.

Second, SigLIP achieves the strongest overall retrieval performance. On Oxford5k/Paris6k, it obtains the highest mAP and Top-1 Accuracy among the evaluated methods. This is likely because SigLIP benefits from large-scale vision-language pretraining, allowing it to learn more general and transferable visual representations.

Third, frozen ImageNet CNNs are useful but limited. EfficientNet-B0 performs best among the CNN backbones, while ResNet50 and VGG19 are weaker. However, all frozen CNN embeddings remain below SigLIP because ImageNet classification pretraining is not directly optimized for instance-level landmark retrieval. The speed-accuracy plots also show that CNN embedding variants form a middle region: they are much stronger than traditional baselines while keeping latency moderate.

9 Conclusion and Future Work

A. Text-to-Text Matching

Text-to-text similarity matching shows a clear evolution from surface-level comparison to semantic vector comparison. Classical methods are fast and interpretable, but they struggle with synonymy, polysemy, and context. Modern sentence-vector methods based on LLMs provide stronger semantic understanding and better practical performance on STSb. Future work should test more sentence-transformer models, compare bi-encoders with cross-encoders, and evaluate domain-specific fine-tuning.

B. Image-to-Image Matching

Image-to-image matching shows a clear gap between traditional descriptors and learned embeddings. Hand-crafted methods are useful as lightweight baselines, but they struggle with viewpoint changes, lighting variation, background clutter, and semantic differences. CNN embeddings provide a strong middle ground, especially EfficientNet-B0, while SigLIP achieves the best overall accuracy. Future work should add qualitative retrieval examples, retrieval-specific fine-tuning, multi-scale inference, whitening, query expansion, and re-ranking to improve instance-level retrieval.

10 References

References

- [1] *A Survey of Text-Matching Techniques*. Survey paper referenced for text similarity method taxonomy.
- [2] Mathilde Caron et al. “Emerging Properties in Self-Supervised Vision Transformers.” In: *International Conference on Computer Vision*. 2021.
- [3] Navneet Dalal and Bill Triggs. “Histograms of Oriented Gradients for Human Detection.” In: *IEEE Conference on Computer Vision and Pattern Recognition*. 2005.
- [4] Alex Krizhevsky. *Learning Multiple Layers of Features from Tiny Images*. Tech. rep. 2009.
- [5] David G. Lowe. “Distinctive Image Features from Scale-Invariant Keypoints.” In: *International Journal of Computer Vision* (2004).
- [6] Filip Radenović, Giorgos Tolias, and Ondřej Chum. “Fine-Tuning CNN Image Retrieval with No Human Annotation.” In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2018).
- [7] Alec Radford et al. “Learning Transferable Visual Models From Natural Language Supervision.” In: *International Conference on Machine Learning*. 2021.

A Appendix

A. Text-to-Text Matching Supplement

Additional text experiments can include full correlation tables, example sentence pairs, and error cases where lexical overlap is misleading or where semantic models fail because of domain-specific meaning.

B. Image-to-Image Matching Supplement

Additional image experiments can include qualitative top-k retrieval examples, SIFT key-point match visualizations, embedding-space projections using t-SNE or UMAP, and full memory/latency logs for each method.